Exchange rate and Production network

Bonn Macro Internal Seminar

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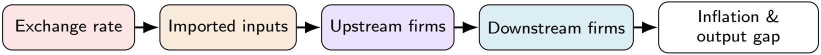
Motivation



Firms buy inputs mostly from *other domestic firms*, not just labor — a cost shock to one sector mechanically hits everyone downstream

Imported inputs are the norm (Chile: up to 19.5% in Manufacturing) — so an FX move is, for the firm, a domestic cost shock

Network models get this propagation right but ignore the exchange rate; open-economy models get the exchange rate right but ignore the network

Does accounting for production networks change the optimal exchange-rate regime?

Literature

Network side (closed economy): Rubbo (2024); La'O & Tahbaz-Salehi (2022); Pasten, Schoenle & Weber (2020); Ozdagli & Weber (2026); Ghassibe (2021)

Open-economy side (no network): Galí & Monacelli (2005); Fanelli & Straub (2021); Schmitt-Grohé & Uribe (2003); Gopinath & Itskhoki (2022); Amiti et al. (2019); Auer, Levchenko & Sauré (2019); Itskhoki & Mukhin (2023); Bianchi & Coulibaly (2025); Coulibaly (2023)

Open economy with network:

- **Gnocato, Montes-Galdón & Stamato (2025, ECB); Kalemli-Özcan et al. (2025)** — open economy *with* network (tariffs), fixed Taylor rule, no FX-regime choice[1]
- **Qiu, Wang, Xu & Zanetti (2026, JME)** — closest paper: derives the open-economy DC index directly extending Rubbo, WIOD 43-country calibration; but *static* (no NFA/persistence), no UIP, and no FX-regime choice

Gap: network + open economy + FX regime choice, together.

[1] Details & what these papers cite for open-economy production networks: Appendix.

This Paper

Extend Rubbo (2024) $\rightarrow$ SOE: imported inputs, UIP and debt-elastic premium

Three literatures, none overlapping fully (network only; open economy only; network + open economy but no regime choice) — **first to put all three together**

Calibrate: Chile's real IO table, also Korea, Czechia

Ask: rank FX regimes? How does the network amplify foreign shocks? How do sector heterogeneity and network structure matter?

Result: **Managed** $<$ **Float** $\ll$ **Peg** (welfare loss $10.96/25.15/90.45 \times 10^{-4}$)

Households

Representative household chooses $\{C_t, L_t, B_t, B_t^*\}$ to maximize lifetime utility, subject to a budget constraint with a domestic bond B_t and a foreign bond B_t^*:

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right]$$

subject to

$$P_t^C C_t + B_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i_{t-1}) B_{t-1} + (1+i^*) \left[1 - \psi(\hat{b}_{t-1}^* - \bar{b}^*)\right] S_t B_{t-1}^*$$

Consumption is CES across home/foreign goods, Cobb-Douglas across domestic sectors:

$$C_t = \left[\omega^{1/\eta} (C_t^H)^{\frac{\eta-1}{\eta}} + (1-\omega)^{1/\eta} (C_t^F)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \qquad C_t^H = \prod_i (C_{it}^H)^{\beta_i^H}$$

Firms: Technology

Firms in sector i share Cobb-Douglas technology (labor, domestic inputs, imports) and price under monopolistic competition:

$$Y_{ift} = A_{it}\, L_{ift}^{\alpha_i} \prod_j X_{ijft}^{\Omega_{ij}^H}\, M_{ift}^{\Omega_i^F}$$

$$MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} \, (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

Two cost-push channels: MC_{it} moves with A_{it} (TFP) *and* S_t, the exchange rate (via the import cost $S_t P_t^{F*}$).

The theory holds for any number of sectors N; the quantitative results use $N = 3$ (Resource, Manufacturing, Services), collapsed from Chile's 12-sector national IO table.

Firms: Price Setting

Calvo pricing through the network delivers the (vector) Phillips curve:

$$\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}_t\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t + \Gamma\Delta e_t$$

Same amplifier, two shocks: $\mathcal{B}, \Gamma$ are the *same* rigidity-adjusted operator $\tilde{A} \equiv \Delta(I - \Omega^H\Delta)^{-1}$, applied to labor share $\boldsymbol{\alpha}$ vs. import share $\boldsymbol{\omega}_F$:

$$\mathcal{B} = \frac{\tilde{A}\boldsymbol{\alpha}}{\kappa}, \qquad \Gamma = \frac{\tilde{A}\boldsymbol{\omega}_F}{\kappa}$$

where:

$\boldsymbol{\pi}_t, \tilde{y}_t$: infl., gap	$\boldsymbol{\chi}_t$: TFP cost-push	Δe_t: exch. rate chg.	κ: GE scalar
$\mathcal{B}, \Gamma$: PC slope, FX pass-thru	$\mathcal{V}$: TFP cost-push matrix	$\tilde{A}$: rigidity-adj. Leontief	

Firms: Production Network

The **Leontief inverse** sums direct and indirect linkages:

$$(I - \Omega^H)^{-1} = I + \Omega^H + (\Omega^H)^2 + \cdots$$

applied to consumption shares β^H (Households) it gives:

$$\boldsymbol{\lambda}_D^\top = (\boldsymbol{\beta}^H)^\top (I - \Omega^H)^{-1}, \qquad \mathcal{M} = (I - \Omega^H)^{-1}\boldsymbol{\omega}_F, \qquad \mathcal{M}^X = (I - \Omega^H)^{-1}\boldsymbol{\omega}_X$$

where:

- $\lambda_{D,i}$: **Domar weight** — sector i's total weight in final demand, network-adjusted (can exceed 1 for downstream sectors)
- $\mathcal{M}_i$: **Import centrality** — sector i's total (direct + indirect) exposure to import costs
- $\mathcal{M}_i^X$: **Export centrality** — sector i's total (direct + indirect) exposure to export demand

Foreign Sector

Law of one price and UIP with a debt-elastic NFA stabilizer $\psi(\cdot)$ and an exogenous risk-premium shock RP_t:

$$P_t^{FH} = S_t P_t^{F*}, \qquad i_t = i^* + \mathbb{E}_t \Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t$$

NFA accumulates the trade balance:

$$\hat{b}_t^* = \frac{1 + i_{t-1}}{\Pi_t^C} \hat{b}_{t-1}^* + \frac{P_t^H \sum_i EX_{it} - P_t^{FH} IM_t}{P_t^C}, \qquad EX_{it} = \kappa_{EX,i} D_t^* \, PX_t \left(\frac{P_{it}}{P_t^{FH}} \right)^{-\theta^*}$$

Two channels to the domestic economy:

$$\hat{s}_t + \hat{p}_t^{F*} \;\to\; MC_{it} \;\to\; \pi_{it} \quad \text{(cost-push)} \qquad \hat{s}_t \;\to\; EX_{it} \;\to\; \tilde{y}_t \;\to\; \pi_{it} \quad \text{(demand)}$$

where:

P_t^{F*}: foreign price of imports	RP_t: risk-premium shock	D_t^*: foreign demand shock	PX_t: export price/ToT shock
$\psi(\cdot)$: debt-elastic NFA stabilizer	$\hat{b}_t^*$: NFA (% GDP)	$\kappa_{EX,i}$: export scale, sector i	θ^*: export demand elasticity
$\hat{s}_t \equiv \log(S_t/\bar{S})$: log exch. rate	$\Delta e_{t+1} \equiv \mathbb{E}_t \hat{s}_{t+1} - \hat{s}_t$: exp. depreciation	(hats: log-deviation from steady state, throughout)	

Monetary Policy Regimes

Free float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$$

The exchange rate adjusts endogenously.

Hard peg

$$\hat{s}_t = 0$$

Adjustment occurs through domestic inflation and output.

Managed float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \hat{s}_t$$

The central bank partially stabilizes the exchange rate.

Equilibrium System

Phillips curve:

$$\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}_t\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t + \Gamma\Delta e_t$$

IS curve:

$$\tilde{y}_t = \mathbb{E}_t\tilde{y}_{t+1} - \frac{1}{\gamma}\big(i_t - \mathbb{E}_t\pi_t^C - r_t^{\mathrm{nat}}\big) - \frac{\eta(1-\omega)}{\gamma}\mathbb{E}_t\Delta\big(\hat{s}_t + \hat{p}_t^{F*} - \hat{p}_t^H\big)$$

Foreign sector:

$$i_t = i^* + \mathbb{E}_t\Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t$$

Policy rule:

$$\hat{i}_t = \begin{cases} \phi_\pi\pi_t^{DC} + \phi_y\tilde{y}_t & \text{Float} \\ \text{residual } (\hat{s}_t = 0) & \text{Peg} \\ \phi_\pi\pi_t^{DC} + \phi_y\tilde{y}_t + \phi_s\hat{s}_t & \text{Managed} \end{cases}$$

Seven AR(1) shocks (1% s.d. each): sector TFP A_1, A_2, A_3 (ρ=0.90), import price PF (ρ=0.85), foreign demand D^* (ρ=0.80), export price/ToT PX (ρ=0.85), risk premium RP (ρ=0.80).

Calibration: Chile Data

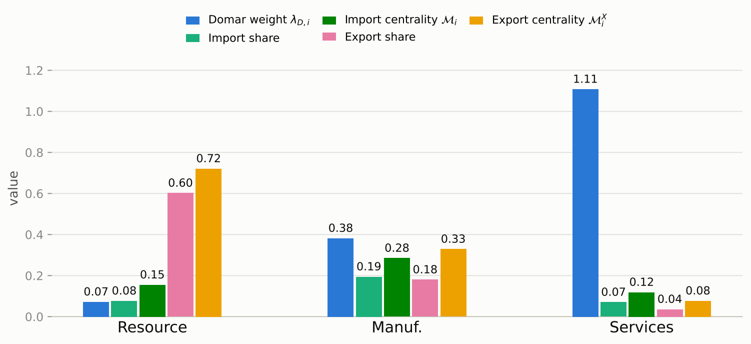
Param.	Description	Value	Source
$\alpha_1, \alpha_2, \alpha_3$	Labor share	0.503, 0.359, 0.604	Data: BCCh IO
$\Omega_1^F, \Omega_2^F, \Omega_3^F$	Import cost share	0.077, 0.195, 0.070	
$\beta_1^H, \beta_2^H, \beta_3^H$	HH consumption share	0.027, 0.229, 0.744	
Export share$_{1,2,3}$	Export share of output	0.602, 0.180, 0.036	
$\delta_1, \delta_2, \delta_3$	Calvo reset prob.	0.90, 0.31, 0.16	Lit.: Dhyne et al. (2005); N-S (2008)
ψ	Risk-premium elasticity	0.020	Key mechanism param. (see appendix)
ϕ_π	Taylor: inflation resp.	1.50	
ϕ_y	Taylor: output-gap resp.	0.50	
ϕ_s	Taylor: FX-smoothing resp.	0.30	

Ω^H (order: Res., Man., Serv.; source: BCCh IO table):

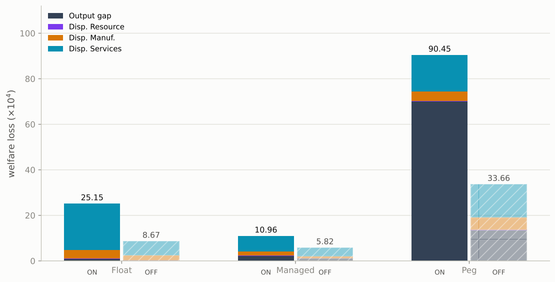
$$\Omega^H = \begin{pmatrix} 0.075 & 0.153 & 0.193 \\ 0.099 & 0.202 & 0.145 \\ 0.002 & 0.058 & 0.266 \end{pmatrix}$$

Standard literature values (not paper-specific; sources in Appendix): β=0.99, γ=1.00, φ=2.00, η=1.50, ε=8.00, θ^*=2.00, $\beta^{F,tot}$=0.10.

Network Properties: Import & Export Exposure



Welfare Ranking: Managed Float Dominates



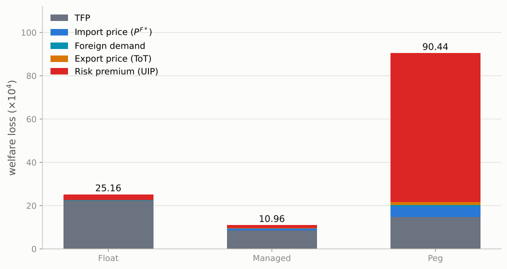
Managed $<$ **Float** $\ll$ **Peg:** 10.96/25.15/90.45 ($\times 10^{-4}$)

Float & Managed: almost entirely price dispersion, concentrated in Services

Peg: almost entirely output gap

Without the network: **same ranking**, but the network roughly **doubles-to-triples** every regime's loss

What Drives Each Regime's Loss? Shock Decomposition



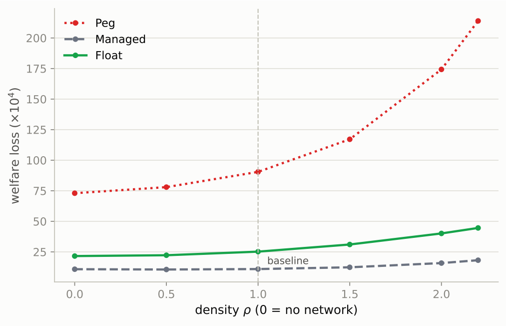
Peg: risk-premium/UIP $\rightarrow$ 76% of loss, mostly output gap

Float & Managed: TFP shocks dominate instead — risk-premium barely registers since the exchange rate (Float) or partial smoothing (Managed) absorbs it before it reaches output

Import price, foreign demand, and export price are second-order for every regime

Isolating the Network Channel

Does the network's effect scale *continuously* with how interconnected the economy is?



Denser network amplifies shock propagation for *everyone*, but **Peg** has no policy lever ($\phi_\pi, \phi_y = 0$) to lean against it, so the amplification passes straight into welfare loss; **Float**/**Managed** actively damp part of it with the interest rate — **gap widens** as $\rho \uparrow$

How the Network Amplifies Shocks

The network **dampens** the inflation peak but **amplifies** the output-gap peak — in *every* regime, for *both* shocks.

Shock	Regime	Inflation peak	Output-gap peak
		(Network ON ÷ Network OFF)	
Import-price	Float	0.40×	2.77×
	Managed	0.61×	2.28×
	Peg	0.86×	**3.15×**
Risk-premium	Float	0.41×	**2.57×**
	Managed	0.57×	0.98×
	Peg	0.67×	2.16×

Note: $< 1\times$ = network shrinks the peak; $> 1\times$ = network amplifies it. (Peak IRF response over 20 quarters; exchange-rate peak itself barely moves with the network)

Network Exposure & Sector Welfare Preferences

Each cell: **Network ON / Network OFF**.

Price-dispersion cost ($\times 10^4$)

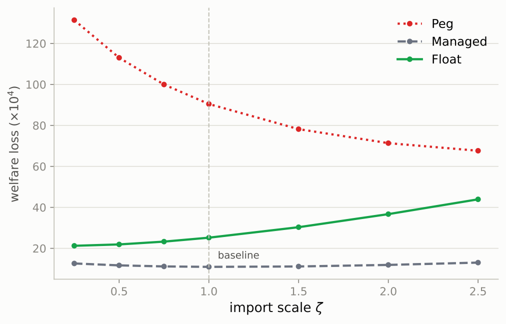
Sector	**Float**	**Managed**	**Peg**
Resource	0.05 / 0.03	0.04 / 0.02	0.27 / 0.27
Manuf.	3.87 / 2.23	1.92 / 0.94	4.28 / 5.32
Services	**20.25** / 6.22	6.71 / 3.74	**15.88** / 14.59

Var(own output gap) ($\times 10^4$)

Sector	**Float**	**Managed**	**Peg**
Resource	0.13 / 0.02	0.14 / 0.04	0.20 / 0.12
Manuf.	0.58 / 0.16	0.38 / 0.08	1.98 / 0.23
Services	0.38 / 0.15	2.31 / 1.33	**35.16** / 8.97

Every sector agrees: Managed $<$ **Float** $<$ **Peg** — Services' huge stake in avoiding Peg is mostly **network-driven** (output-gap cost 8.97 $\rightarrow$ 35.16 with the network on).

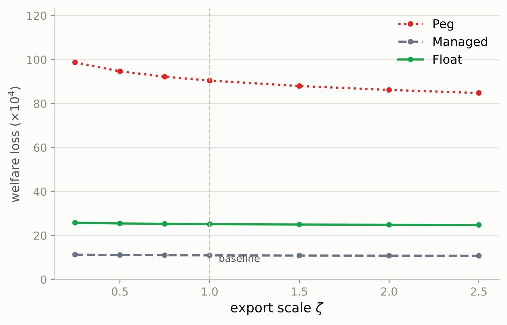
Import Openness



Managed $<$ Float $<$ Peg robust throughout

More import exposure means more $\Gamma \Delta e_t$ cost-push for **Float** (loss rises), but that channel is structurally zero under **Peg** ($\hat{s}_t \equiv 0$), so its loss instead falls via the labor-share renormalization — **gap converges**

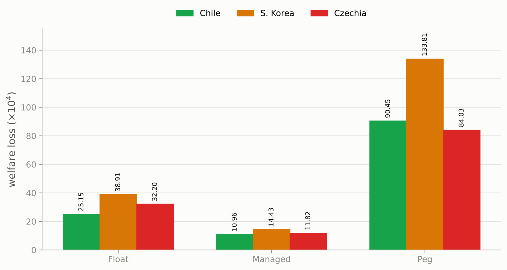
Export Openness



Managed $<$ Float $<$ Peg robust throughout

Exports work only through **demand** ($EX_{it} \rightarrow \tilde{y}_t$), never marginal cost — a pure buffer, so **all regimes improve**, and **Peg gains most** since it has no FX buffer of its own

Robustness: South Korea & Czechia



Managed $<$ Float $\ll$ Peg in every calibration

Korea higher throughout: denser network & higher import exposure

Conclusion

The network matters for severity and incidence, not the ranking: switching it on raises welfare losses in every regime and reallocates who bears them, but **Managed $<$ Float $<$ Peg either way**

Managed float wins because it dampens the FX pass-through channel without fully reimposing peg's loss of autonomy

Robust across three real-data calibrations (Chile, Korea, Czechia) and a genuine second-order solution